Lesson Planning

Complete Lesson Plan Book

Class: Class 5

Subject: Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: The Human Body: Major Organs

Unit 2: The Skeletal and Muscular System

Unit 3: Plants: Parts and Their Functions

Unit 4: Matter: Solids, Liquids and Gases

Unit 5: Force, Motion and Friction

Unit 6: Light and Shadows

Unit 7: Sound: Production and Travel

Unit 8: The Solar System

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Unit 1: The Human Body: Major Organs

Learning Objectives

• Students will be able to name the major organs: heart, lungs, stomach, liver and kidneys.

• Students will be able to state one main function of each major organ.

• Students will be able to locate the organs on a simple body diagram.

Key Concepts

• The heart pumps blood around the body.

• The lungs take in oxygen and release carbon dioxide.

• The stomach digests food.

• The kidneys clean the blood and make urine.

Teaching Methodology (Step by Step)

1. Begin with a 5-minute warm-up: ask 2-3 oral questions revising the previous lesson and linking it to The Human Body: Major Organs.

2. Present the key ideas of The Human Body: Major Organs on the board with one clear example or a labelled diagram.

3. Explain each point slowly in simple words, asking a short check-question after every point.

4. Do guided practice: solve 3-4 examples of The Human Body: Major Organs together, inviting students to complete steps on the board.

5. Close with a 5-minute oral recap in which students state one new thing they learned about The Human Body: Major Organs.

Classroom Activities

• Body map: students label major organs on a large outline diagram in groups.

• Heartbeat check: students feel their pulse before and after 20 jumps.

Assessment Questions

1. Name any four major organs of the human body.

2. What is the main function of the lungs?

3. Which organ digests our food?

Homework

• Draw the human body outline and label five major organs.

Time Allocation

Two periods (80 minutes): 10 min warm-up and revision, 25 min concept presentation, 25 min guided and independent practice, 15 min activity and assessment, 5 min closure and homework.

Unit 2: The Skeletal and Muscular System

Learning Objectives

• Students will be able to state that the skeleton gives shape and protection to the body.

• Students will be able to name major bones: skull, ribs, backbone, arm and leg bones.

• Students will be able to explain that muscles help bones move.

Key Concepts

• The skeleton is made of bones; an adult has 206 bones.

• The skull protects the brain; ribs protect heart and lungs.

• Muscles are attached to bones and pull them to move.

• Joints: places where two bones meet.

Teaching Methodology (Step by Step)

1. Start with rapid-fire mental revision: 5 quick questions from the last lesson connected to The Skeletal and Muscular System.

2. Demonstrate The Skeletal and Muscular System step by step on the board while students watch, listen and copy.

3. Move to guided practice: students attempt 2-3 similar items with the teacher's support.

4. Give independent practice: a short written task on The Skeletal and Muscular System completed individually in notebooks.

5. Check a few notebooks on the spot, correct common mistakes on the board and summarise The Skeletal and Muscular System.

Classroom Activities

• Bone count: students touch and name bones in their own arms and legs.

• Model joint: make a simple elbow joint with cardboard strips and a split pin.

Assessment Questions

1. How does the skeleton protect the body? Give two examples.

2. What is a joint?

3. Name three bones of the human body.

Homework

• List five activities and the muscles or joints they use.

Time Allocation

Two periods (80 minutes): 10 min revision, 20 min explanation with examples, 30 min practice and class activity, 15 min oral and written check, 5 min homework briefing.

Unit 3: Plants: Parts and Their Functions

Learning Objectives

• Students will be able to name the parts of a plant: root, stem, leaf, flower and fruit.

• Students will be able to state the function of each part.

• Students will be able to identify parts on a real plant.

Key Concepts

• Roots absorb water and hold the plant in soil.

• The stem carries water to leaves and supports the plant.

• Leaves make food for the plant.

• Flowers grow into fruits which hold seeds.

Teaching Methodology (Step by Step)

1. Open with a hook: a short story, riddle or real-life example that introduces Plants: Parts and Their Functions.

2. Lead a class discussion on Plants: Parts and Their Functions with 4-5 guiding questions; record key answers on the board.

3. Organise the main activity in pairs or small groups according to the plan for Plants: Parts and Their Functions.

4. Let groups share their work; add corrections and the important points of Plants: Parts and Their Functions.

5. Finish with a quick oral or written check on Plants: Parts and Their Functions and announce the homework.

Classroom Activities

• Plant dissection: groups separate a real plant into parts and label them.

• Leaf prints: make prints of different leaves and name the plant if known.

Assessment Questions

1. Which part of the plant absorbs water from the soil?

2. What is the function of leaves?

3. Draw a plant and label four parts.

Homework

• Observe a plant at home and write the function of each visible part.

Time Allocation

Two periods (80 minutes): 10 min warm-up and revision, 25 min concept presentation, 25 min guided and independent practice, 15 min activity and assessment, 5 min closure and homework.

Unit 4: Matter: Solids, Liquids and Gases

Learning Objectives

• Students will be able to classify objects as solids, liquids or gases.

• Students will be able to state two properties of each state of matter.

• Students will be able to give daily-life examples of each state.

Key Concepts

• Solids have fixed shape and volume.

• Liquids take the shape of their container.

• Gases spread to fill all available space.

• Examples: stone, water, air.

Teaching Methodology (Step by Step)

1. Begin with 5 minutes of silent or loud reading connected to Matter: Solids, Liquids and Gases.

2. Explain the new words and key ideas of Matter: Solids, Liquids and Gases; write difficult words with meanings on the board.

3. Read the passage or examples of Matter: Solids, Liquids and Gases together, with students taking turns.

4. Students complete a short written exercise on Matter: Solids, Liquids and Gases in their notebooks.

5. Review the answers together, praise good work and give a brief summary of Matter: Solids, Liquids and Gases.

Classroom Activities

• Sorting station: students sort 12 objects or pictures into solid, liquid and gas groups.

• Balloon demo: inflate a balloon to show air (gas) takes up space.

Assessment Questions

1. Give two properties of liquids.

2. Is smoke a solid, liquid or gas? Explain.

3. Name two solids and two gases from your classroom.

Homework

• List 5 solids, 5 liquids and 3 gases found at home.

Time Allocation

Two periods (80 minutes): 10 min revision, 20 min explanation with examples, 30 min practice and class activity, 15 min oral and written check, 5 min homework briefing.

Unit 5: Force, Motion and Friction

Learning Objectives

• Students will be able to define force as a push or a pull.

• Students will be able to identify friction as a force that slows motion.

• Students will be able to give examples of motion in daily life.

Key Concepts

• Force: push or pull acting on an object.

• Friction opposes motion; rough surfaces make more friction.

• Motion: change of position with time.

• Gravity pulls objects towards the Earth.

Teaching Methodology (Step by Step)

1. Begin with a 5-minute warm-up: ask 2-3 oral questions revising the previous lesson and linking it to Force, Motion and Friction.

2. Present the key ideas of Force, Motion and Friction on the board with one clear example or a labelled diagram.

3. Explain each point slowly in simple words, asking a short check-question after every point.

4. Do guided practice: solve 3-4 examples of Force, Motion and Friction together, inviting students to complete steps on the board.

5. Close with a 5-minute oral recap in which students state one new thing they learned about Force, Motion and Friction.

Classroom Activities

• Ramp race: roll a toy car on smooth and rough surfaces; compare distances.

• Tug of war: feel push and pull forces in a short class game.

Assessment Questions

1. What is a force? Give one example of a push and one of a pull.

2. Why does a ball stop rolling on grass but keep rolling on a smooth floor?

3. Name the force that pulls a dropped pencil to the ground.

Homework

• Write 5 examples of friction helping us in daily life.

Time Allocation

Two periods (80 minutes): 10 min warm-up and revision, 25 min concept presentation, 25 min guided and independent practice, 15 min activity and assessment, 5 min closure and homework.

Unit 6: Light and Shadows

Learning Objectives

• Students will be able to state that light travels in a straight line.

• Students will be able to explain how shadows are formed.

• Students will be able to classify materials as transparent, translucent or opaque.

Key Concepts

• Light comes from sources like the sun and bulbs.

• Shadows form when an opaque object blocks light.

• Transparent: glass; translucent: frosted glass; opaque: wood.

• Longer shadows form when light is low.

Teaching Methodology (Step by Step)

1. Start with rapid-fire mental revision: 5 quick questions from the last lesson connected to Light and Shadows.

2. Demonstrate Light and Shadows step by step on the board while students watch, listen and copy.

3. Move to guided practice: students attempt 2-3 similar items with the teacher's support.

4. Give independent practice: a short written task on Light and Shadows completed individually in notebooks.

5. Check a few notebooks on the spot, correct common mistakes on the board and summarise Light and Shadows.

Classroom Activities

• Shadow puppets: students make hand shadows with a torch in a darkened corner.

• Material test: shine a torch through glass, paper and cardboard; record results.

Assessment Questions

1. How is a shadow formed?

2. Classify: window glass, notebook, tracing paper.

3. Why are shadows longer in the morning?

Homework

• Observe your shadow at three times of day and draw its length.

Time Allocation

Two periods (80 minutes): 10 min revision, 20 min explanation with examples, 30 min practice and class activity, 15 min oral and written check, 5 min homework briefing.

Unit 7: Sound: Production and Travel

Learning Objectives

• Students will be able to state that sound is produced by vibrations.

• Students will be able to explain that sound travels through air.

• Students will be able to compare loud and soft sounds.

Key Concepts

• Vibrating objects produce sound.

• Sound travels in waves through air, water and solids.

• Loud sounds have bigger vibrations.

• The ear receives sound vibrations.

Teaching Methodology (Step by Step)

1. Open with a hook: a short story, riddle or real-life example that introduces Sound: Production and Travel.

2. Lead a class discussion on Sound: Production and Travel with 4-5 guiding questions; record key answers on the board.

3. Organise the main activity in pairs or small groups according to the plan for Sound: Production and Travel.

4. Let groups share their work; add corrections and the important points of Sound: Production and Travel.

5. Finish with a quick oral or written check on Sound: Production and Travel and announce the homework.

Classroom Activities

• Rubber-band guitar: pluck bands of different tightness; hear pitch change.

• String telephone: talk through two cups joined by string.

Assessment Questions

1. How is sound produced? Give an example.

2. Can sound travel in empty space? Why?

3. Name two materials sound can travel through.

Homework

• List 5 sounds heard at home and say whether each is loud or soft.

Time Allocation

Two periods (80 minutes): 10 min warm-up and revision, 25 min concept presentation, 25 min guided and independent practice, 15 min activity and assessment, 5 min closure and homework.

Unit 8: The Solar System

Learning Objectives

• Students will be able to name the eight planets in order from the sun.

• Students will be able to state that the Earth revolves around the sun.

• Students will be able to describe the sun as a star and the moon as a satellite.

Key Concepts

• The sun is a star at the centre of the solar system.

• Eight planets: Mercury to Neptune.

• The Earth rotates (day and night) and revolves (year).

• The moon revolves around the Earth.

Teaching Methodology (Step by Step)

1. Begin with 5 minutes of silent or loud reading connected to The Solar System.

2. Explain the new words and key ideas of The Solar System; write difficult words with meanings on the board.

3. Read the passage or examples of The Solar System together, with students taking turns.

4. Students complete a short written exercise on The Solar System in their notebooks.

5. Review the answers together, praise good work and give a brief summary of The Solar System.

Classroom Activities

• Planet parade: 8 students hold planet cards and stand in order from the sun.

• Model making: groups arrange balls in orbit order on a chart.

Assessment Questions

1. Name the eight planets in order.

2. What causes day and night on Earth?

3. Is the sun a star or a planet? Explain.

Homework

• Draw the solar system and label all eight planets.

Time Allocation

Two periods (80 minutes): 10 min revision, 20 min explanation with examples, 30 min practice and class activity, 15 min oral and written check, 5 min homework briefing.